Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHNOLOGY

(Autonomous Institute Affiliated to University of Mumbai)

Munshi Nagar, Andheri (W), Mumbai – 400 058.

End Semester Examination

Dec 2025

Max. Marks: 100 | **Duration: 3 Hours.**

Class: Third Year BTech(CE/CSE) | **Semester: V**

Course Code: CE305/CS305 | **Program: BTech(CE/CSE)**

Name of the Course: Cryptography and Network Security

Instruction:

1) **All questions are compulsory.**
2) **Draw neat diagrams and write equations wherever necessary.**
3) **Assume suitable data if necessary but justify the same.**

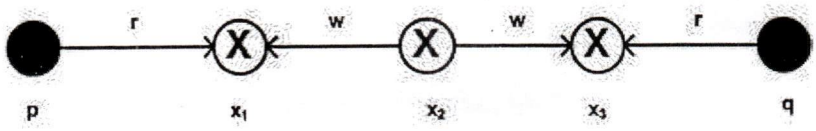
Course Outcomes: At the end of the course, student will be able to:

1) **Understand the fundamental principles of security and cryptography e.g. number theory, finite fields etc.**
2) **Make use of classical and modern encryption and decryption algorithms to secure data, ensuring confidentiality, authenticity, and efficiency in cryptographic systems.**
3) **Develop a deep understanding of cryptographic data integrity algorithms, including hash functions, message authentication codes (MACs), and digital signatures.**
4) **Apply key management, authentication mechanisms, intrusion detection systems (IDS), and firewalls to enhance security, protect data, and mitigate cyber threats.**
5) **Understand and implement PGP, SSL, TLS, and IPSec protocols to enhance secure communication, ensure data integrity, and support authentication in networked systems.**

Q. No.	Question	Max. Marks	CO-BL
Q. 1 A	Numerical Theorist Fermat and Rabin Miller got into an argument over number 294409. They both went on checking whether this number is prime or composite using their own method. What result will they get if they use base 2? Can we say that 294409 Carmichael number?	10	CO1-3
Q. 1 B	Two mathematicians are debating about the algebraic structure of the set $\mathbb{Z}_6 = \{0,1,2,3,4,5\}$, where addition and multiplication are defined modulo 6. • Mathematician A claims that $\mathbb{Z}_6$ is a field. • Mathematician B claims that $\mathbb{Z}_6$ is only a ring and not a field. Both approach you to settle the argument. Your task is to prove step by step whether $\mathbb{Z}_6$ is a *ring*, and whether it satisfies the additional properties required to be a *field*. In your proof: 1. Verify each ring property (closure, associativity, identity elements, additive inverses, distributive law, etc.). 2. Then test the field property — whether every nonzero element in $\mathbb{Z}_6$ has a multiplicative inverse. Based on your results, clearly conclude whether $\mathbb{Z}_6$ is only a ring or a field.	10	CO1-3
Q. 2 A	A generalization of the Caesar cipher, known as the affine Caesar cipher, has the following form. For each plaintext letter p, substitute the ciphertext letter C: $C = E([a,b],p) = (ap + b) mod 26$ For Decryption we have: $p = D([a,b],C) = a^{-1}(p - b) mod 26$ A basic requirement of any encryption algorithm is that it be one-to-one. That is, if $p \neq q$, then $E(k,p) \neq E(k,q)$. Otherwise, decryption is impossible, because more than one plaintext character maps into the same ciphertext character. The affine Caesar cipher is not one-to-one for all values of a. For example, for $a = 2$ and $b = 3$, then $E([a,b],0) = E([a,b],13) = 3$.	10	CO2-4

<table>
<tr><td></td><td>i) Are there any limitations on the value of b? Explain why or why not. [2 M]
ii) Determine which values of a are not allowed and which are not allowed. Justify your answer. [5 M]
iii). For $a = 7$and $b = 3$, encrypt the plaintext "HELP" using the affine cipher. Decrypt the resulting ciphertext to verify your answer. [3 M]</td><td></td><td></td></tr>
<tr><td>Q. 2 B</td><td>Consider AES Encryption Algorithm. Perform all four transformations (namely subBytes, shiftRows, MixColumns and AddRoundKey in correct sequence) on the output state (in Hex) of 2nd Round with the Round-2 Key (in Hex) as shown below:
<table><tr><th colspan="4">Output of 2nd Round</th></tr><tr><td>89</td><td>6A</td><td>0C</td><td>A5</td></tr><tr><td>2B</td><td>7E</td><td>15</td><td>16</td></tr><tr><td>28</td><td>AE</td><td>D2</td><td>A6</td></tr><tr><td>AB</td><td>F7</td><td>97</td><td>75</td></tr></table><table><tr><th colspan="4">Round 2 key</th></tr><tr><td>D6</td><td>AA</td><td>74</td><td>FD</td></tr><tr><td>D2</td><td>AF</td><td>72</td><td>FA</td></tr><tr><td>DA</td><td>A6</td><td>78</td><td>F1</td></tr><tr><td>D6</td><td>AB</td><td>76</td><td>FE</td></tr></table>Show the output of 3rd round. Note that AES s-box mapping is attached at the end.</td><td>10</td><td>CO3-3</td></tr>
<tr><td>Q. 3 A</td><td>Consider SHA-512 hash function. Find the output of next round (i+1)th round in hexadecimal format only, if the key K_i = 0x428a2f98d728ae22 and message part W_i =0x0000000000000000. Following table shows output of ith round of SHA-512.
<table><tr><th>Variable</th><th>Value (Hex)</th></tr><tr><td>a</td><td>0x6a09e667f3bcc908</td></tr><tr><td>b</td><td>0xbb67ae8584caa73b</td></tr><tr><td>c</td><td>0x3c6ef372fe94f82b</td></tr><tr><td>d</td><td>0xa54ff53a5f1d36f1</td></tr></table><table><tr><th>Variable</th><th>Value (Hex)</th></tr><tr><td>e</td><td>0x510e527fade682d1</td></tr><tr><td>f</td><td>0x9b05688c2b3e6c1f</td></tr><tr><td>g</td><td>0x1f83d9abfb41bd6b</td></tr><tr><td>h</td><td>0x5be0cd19137e2179</td></tr></table></td><td>10</td><td>CO3-3</td></tr>
<tr><td>Q. 3 B</td><td>Consider the Schnorr Digital Signature Scheme with following parameters: Prime number p=29 and its prime factor q = 7. Alice generates a random integer number a = 16 and thus (a,p,q)=(16,29,7) becomes global public key. Alice chooses a random number s=3 as private key. Alice wants to generate a digital signature for the message M=13 using random number r=2. [Assume hash function H(u||v) = u+v mod 29.]
ii) Show generation of digital signature for M=13 using random number r=2.
iii) Show the verification of generated digital signature.

OR

Consider the TGP Model. Discuss all four graph rewriting rules of TGP Model. The following protection graph consisting of two subjects and three subject-objects. Let p and q be subjects. x_1 to x_3 are subject-objects. Further, the access rights are r(read) and w(write) are also called take(t) and grant(g) rights respectively.
 

Prove that p can take/grant q using repetitive use of TGP rewriting rules. Show the proof using all intermediate protection graphs.</td><td>10</td><td>CO3-4</td></tr>
</table>

Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHNOLOGY

(Autonomous Institute Affiliated to University of Mumbai)

Munshi Nagar, Andheri (W), Mumbai – 400 058.

Q. 4 A	Which vulnerability is found related to User Authentication in the original Needham and Schroeder Key distribution Scheme using KDC and Symmetric Key Distribution? How did this vulnerability overcome in User Authentication Process by Denning's Proposal? Show all messages sent among entities for vulnerability and its solution.	10	CO4-2
Q. 4 B	Describe how Host-Based Intrusion Detection Systems (HIDS) and Network-Based Intrusion Detection Systems (NIDS) operate, and provide appropriate real-life examples for each. Also, differentiate between Signature-based detection and Anomaly-based detection methods, explaining how they detect intrusions, and include one example for each approach.	10	CO4-4
Q. 5 A	You are sending a confidential report through email using PGP. Illustrate the entire process — from message creation to encryption, signing, transmission, and decryption — showing how PGP ensures both privacy and authenticity	10	CO5-2
Q. 5 B	Illustrate the working of IPSec in Transport Mode and Tunnel Mode using suitable diagrams. Clearly differentiate between the two modes in terms of packet structure, header placement, and use cases.	10	CO5-4

AES s-box

x/y	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16